

D678

# Elementary Science Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

In a second-grade unit on animal parts, students in a room with a large class aquarium compare how a duck's webbed feet, a hawk's hooked beak, and a woodpecker's strong beak help each animal get food. The school recently repainted its halls. Which crosscutting concept does the lesson emphasize?

- A. Patterns.
- B. Planning and carrying out investigations.
- C. Systems and system models, because the aquarium is a system.
- D. Structure and function.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

A teacher wants a lesson that highlights the crosscutting concept of stability and change. Which lesson fits?

- A. Students sort leaves by shape and edge.
- B. Students test which ramp height makes a car go farthest.
- C. Students measure the classroom in paper clips and meters.
- D. Students observe a pumpkin decay over four weeks and record what changes and what stays the same.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A fourth-grade team with a strong reputation for hands-on labs is writing an assessment for a standard that asks students to develop a model showing that plants and animals have internal and external structures that support survival. The team has a grant for new microscopes. Which assessment best matches the performance expectation?

- A. Students build and explain a labeled model of an organism, showing how its structures help it survive.
- B. A matching quiz pairing structures with their names.
- C. An essay on why science is important.
- D. Students view cells under the new microscopes and draw what they see.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A fifth-grade class plants a bean seed in a pot holding 500 g of dry soil, placed near a sunny window with a new blue curtain. After eight weeks the plant has gained 40 g of dry mass, while the dry soil has lost less than 1 g. Where did most of the plant's new mass come from?

- A. 500 g of soil nutrients.
- B. Carbon dioxide from the air, used in photosynthesis.
- C. Sunlight, which turns into plant matter.
- D. The soil, which plants eat as food.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

Third graders on a field trip to a pond with a new wooden boardwalk draw a food chain: algae, tadpole, fish, heron. The trip was led by a park ranger with 20 years of experience. Which way should the arrows point?

- A. Both directions between each pair.
- B. From the sun to the heron only.
- C. algae → tadpole → fish → heron
- D. heron → fish → tadpole → algae

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6. [TRANSFER — reversed stem / different variable]

In a food web, an arrow points from grasshopper to frog. What does the arrow show?

- A. The grasshopper eats the frog.
- B. The frog makes its own food from sunlight.
- C. Energy moves from the grasshopper to the frog, because the frog eats the grasshopper.
- D. The grasshopper and frog compete for food.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

Fourth graders visiting a canyon with a new visitor center watch a slow river carry sand and pebbles downstream. Their guide, a retired teacher, points to cracked rocks on the cliff above. Which process is the river showing?

- A. Deposition.
- B. Erosion.
- C. Weathering.
- D. Weathering, because the guide pointed to cracked rocks.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8. [TRANSFER — reversed stem / different variable]

At the mouth of a river, sand and mud pile up to form a delta. Which process formed it?

- A. Weathering.
- B. Deposition.
- C. Erosion.
- D. Volcanic activity.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

A first-grade class tracks the shadow of a flagpole outside their newly built school from morning to afternoon. A student asks why it gets dark each night. The class's favorite book is about the moon. What is the best explanation?

- A. Earth travels around the sun each day.
- B. The sun moves around Earth each day.
- C. Earth spins, so our side turns away from the sun each night.
- D. The moon blocks the sun at night.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**10.**

A student writing in a class journal that the principal reads each week explains: “The arctic fox grew thick fur because it was cold and needed to stay warm.” The class keeps a bearded dragon. How should the teacher correct the idea?

- A.** Foxes with inherited thicker fur survived and had more young, so over many generations thick fur became common.
- B.** All animals change their fur every winter.
- C.** The idea is correct; animals grow what they need.
- D.** Compare with how the bearded dragon grows scales.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

A toy car rolls across a smooth gym floor at the same speed in a straight line. The gym was just refinished and the car is bright red. A fifth grader says a forward force must still be pushing it. Which statement is correct?

- A.** The forces on the car are balanced; no net force is needed to keep it moving steadily.
- B.** No forces at all are acting on it.
- C.** An unbalanced forward force is pushing it.
- D.** Its color adds to its motion.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.** [TRANSFER — reversed stem / different variable]

Which situation shows an unbalanced force?

- A.** A book resting on a desk.
- B.** A car parked on a flat street.
- C.** An elevator moving up at a steady speed.
- D.** A soccer ball curving and slowing as it rolls across grass.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

In a second-grade unit on animal parts, students ~~in a room with a large class aquarium~~ compare how a duck's webbed feet, a hawk's hooked beak, and a woodpecker's strong beak help each animal get food. ~~The school recently repainted its halls.~~ Which crosscutting concept does the lesson emphasize?

**Target word:** which crosscutting concept **Decides:** The shape of a body part relates to what it does.

|   |                   |                                                                  |
|---|-------------------|------------------------------------------------------------------|
| A | Adjacent category | Patterns could appear, but the lesson ties each shape to a job.  |
| B | Confusion set     | A science practice, not a crosscutting concept.                  |
| C | Camouflage        | Keys on the aquarium, which is not the lesson's focus.           |
| D | <b>Correct</b>    | How a part's shape lets it do its job is structure and function. |

## 2. Answer D Concept 1 • transfer

A teacher wants a lesson that highlights the crosscutting concept of stability and change. Which lesson fits?

**Target word:** choose the lesson for a given concept **Decides:** Tracking what stays the same and what changes over time in a system.

|   |                    |                                                                   |
|---|--------------------|-------------------------------------------------------------------|
| A | Confusion set      | Patterns.                                                         |
| B | Adjacent category  | Cause and effect.                                                 |
| C | Inverted operation | Scale, proportion, and quantity.                                  |
| D | <b>Correct</b>     | Change over time in a system is the core of stability and change. |

## 3. Answer A Concept 2

A fourth-grade team ~~with a strong reputation for hands-on labs~~ is writing an assessment for a standard that asks students to develop a model showing that plants and animals have internal and external structures that support survival. ~~The team has a grant for new microscopes.~~ Which assessment best matches the performance expectation?

**Target word:** assessment that fits the performance expectation **Decides:** Students must use the practice (modeling), the core idea, and the crosscutting concept together.

|   |                   |                                                                       |
|---|-------------------|-----------------------------------------------------------------------|
| A | <b>Correct</b>    | Combines the practice, the content, and structure and function.       |
| B | Dropped element   | Drops the modeling practice and the survival explanation.             |
| C | Adjacent category | Writing about science, not the standard's content or practice.        |
| D | Camouflage        | Uses the microscope grant; cells are not the performance expectation. |

#### 4. Answer B Concept 3

A fifth-grade class plants a bean seed in a pot holding 500 g of dry soil, placed near a sunny window ~~with a new blue curtain~~. After eight weeks the plant has gained 40 g of dry mass, while the dry soil has lost less than 1 g. Where did most of the plant's new mass come from?

**Target word:** source of plant mass **Decides:** The soil barely changed; carbon from CO<sub>2</sub> built the sugars.

|          |                   |                                                                                                          |
|----------|-------------------|----------------------------------------------------------------------------------------------------------|
| A        | Wrong stem number | The starting soil mass, which barely changed.                                                            |
| <b>B</b> | <b>Correct</b>    | Plants build sugars from CO <sub>2</sub> and water; carbon from the air makes up most of their dry mass. |
| C        | Adjacent category | Sunlight supplies energy, not mass.                                                                      |
| D        | Confusion set     | The soil lost under 1 g — it cannot account for 40 g.                                                    |

#### 5. Answer C Concept 4

Third graders on a field trip to a pond ~~with a new wooden boardwalk~~ draw a food chain: algae, tadpole, fish, heron. ~~The trip was led by a park ranger with 20 years of experience.~~ Which way should the arrows point?

**Target word:** arrow direction **Decides:** Arrows show energy moving from the eaten to the eater.

|          |                    |                                                                             |
|----------|--------------------|-----------------------------------------------------------------------------|
| A        | Dropped element    | Drops the one-way flow of energy.                                           |
| B        | Adjacent category  | Energy does start with the sun, but it reaches the heron through the chain. |
| <b>C</b> | <b>Correct</b>     | Energy flows from each organism to the one that eats it.                    |
| D        | Inverted operation | Reads the arrow as “eats,” reversing energy flow.                           |

#### 6. Answer C Concept 4 • transfer

In a food web, an arrow points from grasshopper to frog. What does the arrow show?

**Target word:** interpret an arrow — reverse of drawing one **Decides:** Arrows show energy flow to the eater.

|          |                    |                                                                  |
|----------|--------------------|------------------------------------------------------------------|
| A        | Inverted operation | Reads the arrow backward.                                        |
| B        | Adjacent category  | Describes a producer; a frog is a consumer.                      |
| <b>C</b> | <b>Correct</b>     | The arrow points to the eater.                                   |
| D        | Confusion set      | Competition is a different relationship, not shown by one arrow. |

#### 7. Answer B Concept 5

Fourth graders visiting a canyon ~~with a new visitor center~~ watch a slow river carry sand and pebbles downstream. ~~Their guide, a retired teacher,~~ points to cracked rocks on the cliff above. Which process is the river showing?

**Target word:** which process **Decides:** Moving the pieces is erosion.

|          |                   |                                                                           |
|----------|-------------------|---------------------------------------------------------------------------|
| A        | Adjacent category | Deposition is where the river drops the sediment.                         |
| <b>B</b> | <b>Correct</b>    | Water is carrying sediment away.                                          |
| C        | Confusion set     | Weathering breaks rock in place, like the cracked cliff rocks.            |
| D        | Camouflage        | The cracked rocks show weathering, but the question asks about the river. |

### 8. Answer B Concept 5 • transfer

At the mouth of a river, sand and mud pile up to form a delta. Which process formed it?

**Target word:** name the process from the result **Decides:** Sediment dropped in a new place.

|          |                    |                                                                 |
|----------|--------------------|-----------------------------------------------------------------|
| A        | Confusion set      | Breaking rock apart, not piling it up.                          |
| <b>B</b> | <b>Correct</b>     | The slowing river drops sediment, building land.                |
| C        | Inverted operation | Erosion removes and carries sediment; this is where it settles. |
| D        | Adjacent category  | Builds land, but by a different process.                        |

### 9. Answer C Concept 6

A first-grade class tracks the shadow of a flagpole outside their ~~newly built~~ school from morning to afternoon. A student asks why it gets dark each night. ~~The class's favorite book is about the moon.~~ What is the best explanation?

**Target word:** cause of day and night **Decides:** Earth spins on its axis once about every 24 hours.

|          |                    |                                                         |
|----------|--------------------|---------------------------------------------------------|
| A        | Confusion set      | One trip around the sun takes a year.                   |
| B        | Inverted operation | Reverses what is moving.                                |
| <b>C</b> | <b>Correct</b>     | Rotation causes day and night.                          |
| D        | Camouflage         | Keys on the moon; that describes an eclipse, not night. |

### 10. Answer A Concept 7

A student writing in a class journal ~~that the principal reads each week~~ explains: "The arctic fox grew thick fur because it was cold and needed to stay warm." ~~The class keeps a bearded dragon.~~ How should the teacher correct the idea?

**Target word:** correct the adaptation idea **Decides:** Traits spread through populations over generations; individuals do not change by need.

|          |                    |                                                                                        |
|----------|--------------------|----------------------------------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | Adaptation happens to populations through inheritance and survival.                    |
| B        | Absolute language  | Some shed seasonally, but this is not how the trait arose and not true of all animals. |
| C        | Inverted operation | Individuals do not grow inherited traits on demand.                                    |
| D        | Camouflage         | The class pet does not address the misconception.                                      |

### 11. Answer A Concept 8

A toy car rolls across a smooth gym floor at the same speed in a straight line. ~~The gym was just refinished and the car is bright red.~~ A fifth grader says a forward force must still be pushing it. Which statement is correct?

**Target word:** forces on steady motion **Decides:** Constant speed in a straight line means balanced forces.

|          |                   |                                                                             |
|----------|-------------------|-----------------------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | Balanced forces leave motion unchanged.                                     |
| B        | Absolute language | Gravity, the floor's push, and small friction still act; they are balanced. |
| C        | Confusion set     | An unbalanced force would make it speed up.                                 |
| D        | Camouflage        | Irrelevant detail.                                                          |

## 12. Answer D Concept 8 · transfer

Which situation shows an unbalanced force?

**Target word:** choose the unbalanced case **Decides:** A change in speed or direction.

|   |                    |                                                               |
|---|--------------------|---------------------------------------------------------------|
| A | Inverted operation | At rest — balanced.                                           |
| B | Dropped element    | No change in motion.                                          |
| C | Confusion set      | Steady motion — balanced.                                     |
| D | <b>Correct</b>     | Its speed and direction change, so the forces are unbalanced. |

## Triage grid

Write each item number in the box it belongs to.

|                       | <b>Right</b>                                          | <b>Wrong</b>                                                                          |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| <b>Trap</b>        | <b>What it does</b>                                          | <b>In this drill</b> | <b>Your misses</b> |
|--------------------|--------------------------------------------------------------|----------------------|--------------------|
| Dropped element    | Leaves out a required term or step.                          | 3                    |                    |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 1                    |                    |
| Adjacent category  | The right idea in the wrong bucket.                          | 8                    |                    |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7                    |                    |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 2                    |                    |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9                    |                    |
| Camouflage         | Answers a stem fact that was never the question.             | 6                    |                    |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                                    | Rule                                                                                                                                                                                                                      | Trap                                                                                          | Tell                                                                                         |
|---|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1 | <b>Crosscutting concepts</b>                               | Crosscutting concepts are lenses that apply across all sciences: patterns; cause and effect; scale, proportion, and quantity; systems and system models; energy and matter; structure and function; stability and change. | Confusing a crosscutting concept with a practice (what students do) or a core idea (content). | A lesson focus that could apply to any science topic, such as “how shape relates to job.”    |
| 2 | <b>Performance expectations integrate three dimensions</b> | A standard’s performance expectation combines a practice, a core idea, and a crosscutting concept; lessons build toward it, and assessment asks students to use all three together.                                       | Teaching or testing only the content fact.                                                    | An assessment plan for a standard written as “Students who demonstrate understanding can...” |
| 3 | <b>Where a plant’s mass comes from</b>                     | Most of a plant’s added mass comes from carbon dioxide in the air, built into sugars during photosynthesis; soil supplies only small amounts of minerals.                                                                 | Believing plants “eat” soil or get their food from it.                                        | A plant gains mass while the soil’s mass barely changes.                                     |
| 4 | <b>Energy flow in food webs</b>                            | Arrows in a food web point from the organism eaten to the eater, showing the direction energy flows; energy enters from the sun through producers.                                                                        | Drawing arrows from predator to prey (“eats”).                                                | A diagram or question about arrow direction.                                                 |
| 5 | <b>Weathering vs. erosion vs. deposition</b>               | Weathering breaks rock down in place; erosion moves the pieces; deposition drops them in a new place.                                                                                                                     | Using “erosion” for any breaking down of rock.                                                | Whether rock is broken, carried, or dropped.                                                 |
| 6 | <b>Day and night</b>                                       | Day and night are caused by Earth’s rotation on its axis about once every 24 hours; a year is one revolution around the sun.                                                                                              | Saying the sun moves around Earth or that orbiting causes day and night.                      | A daily pattern of sunlight or shadows.                                                      |
| 7 | <b>Adaptation happens to populations</b>                   | Inherited traits that help survival become more common in a population over many generations; individuals do not grow new traits because they need them.                                                                  | “The animal adapted by growing thicker fur because it was cold.”                              | A trait change explained by need or effort.                                                  |
| 8 | <b>Balanced and unbalanced forces</b>                      | Balanced forces leave motion unchanged — at rest or at constant speed in a straight line; only unbalanced forces change speed or direction.                                                                               | Believing a moving object must have an unbalanced force pushing it forward.                   | An object moving at steady speed, or starting, stopping, or turning.                         |

**Course scope (public description):** Elementary Science Curriculum is a comprehensive course that provides an in-depth look at the development and implementation of a robust, inquiry-based science curriculum that aligns

with current educational standards and best practices. Participants explore key concepts in teaching science to young learners, including major concepts within the Earth, life, and physical sciences. The curriculum emphasizes the importance of cross-cutting concepts, science and engineering practices, and disciplinary core ideas in elementary science education to support students' conceptual understanding. By the end of the course, educators are prepared with the foundational knowledge and tools necessary to inspire a lifelong interest in science among their students.